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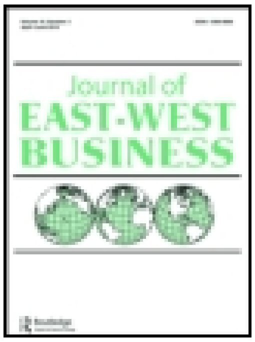

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Startup Success Factors in the Capital Attraction Stage: Founders' Perspective

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ABSTRACT

Only a small percentage of startups attract capital from venture capital investors. To determine the factors which owners of startups consider the most important for attracting seed venture investments, the founders of 40 startups in Latvia and Russia were questioned. The researchers compared organizational and financial factors' importance for two groups of entrepreneurs: those who succeeded and those who failed in attracting funding. The results of the study indicate certain differences between the viewpoints of founders and investors regarding success factors. Based on the factor and regression analysis, the authors developed a model to forecast success in capital attraction.

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Introduction

The challenges faced by startup founders vary at different levels: from idea generation and family support to the creation of market demand. One of the main problems founders of young innovative companies have to solve is the question of financing—its source, conditions, quantity, etc.

According to M. Mason's calculations (Mason 2017), about 137,000 new companies are registered every day worldwide. Entrepreneurial activity is on the rise and startups are popping up at an increased pace. However, only 0.25–2% of all US startup companies seeking venture capital receive financing (Rose 2012). Thus, attracting capital is vital for a young innovative company's survival in the early stages and remains critical for its further expansion in later development stages.

Almost all young innovative companies face difficulties in the capital attraction stage due to the information asymmetry between founders and potential investors (Gompers and Lerner 1999; Davila, Foster, and Gupta 2003), which is most tangible in the period of concept building and

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product or technology development in specialized niches, particularly where investors might not possess the necessary expertise. Still, some companies appear to be more successful in capital attraction than others, which hints at the existence of certain factors, attributed to organizational or financial aspects, which seem to be shared by startup companies and founders that manage to receive investors' trust and seed capital.

Investment at the initial stage can be attracted via different sources of financing: FFF (family, friends, fools), grants, business incubators and accelerators, crowd-founding platforms, business angels, or venture capital investors. The present study focuses on venture capital investors, which are represented by business angels and venture capital funds, which have their own set of requirements which have to be met for the financing to be provided. The majority of studies analyzing the factors most important for early-stage investors mention the leadership skills of the team and particularly the founder as the key aspects for the startup company during the financing round (Rea 1989; Mason 2006; Sudek 2007; Maxwell and Lévesque 2014; Barinova 2015; Bernstein, Korteweg, and Laws 2017).

The research question to be answered in the present article aims to find out which organizational and financial factors have the greatest influence on young innovative enterprises' success in attracting financing at the initial development stage. The analysis is done from the perspective of the startup companies seeking investment, surveying the founders of Latvian and Russian startups to determine the particular factors and factor weights taken into account by business angels and venture capital funds when making decisions regarding investment in young and innovative companies. The main scientific contribution of the present article is that it provides insight into startup founders' views on the kinds of factors that influence capital attraction in the seed stage and the weight assigned to these factors. We believe that this research fills a void in the scientific literature since startup fundraising success factors are evaluated by the entrepreneurs themselves, who have received insufficient attention from leading scientists.

The methods employed to achieve the aim of determining the factors attributed to startups' success in fundraising were processing and analyzing the data obtained from the questionnaires addressing founders of startups, factor analysis, binary logistic regression analysis, and interviews with startup entrepreneurs, business angels and venture capital investors. Additionally, the administrative management theory (Fayol 1917) was tested in the context of startup factors' importance in attracting potential investors.

The present research begins with an overview of factors important for startups' fundraising as disclosed in scientific literature. The literature review is followed by an account of the methodology and a description of the data sources. In the next part, the authors provide an interpretation of the results obtained on factor importance and build a model to forecast the success of a startup in the capital attraction phase. The article ends with conclusions and recommendations for further research.

Literature review

Background of the problem

Innovative development of the economy is one of the main priorities of the European Union, while startups are the companies most likely to come up with innovations (EBRD 2014; European Commission 2014). The rate of survival and success of young innovative companies are critical for the development of new products and creation of new economic sectors (García-Quevedo, Pellegrino, and Vivarelli 2011). It should be mentioned that there is a certain gap between the USA and the European Union with regard to innovative development and productivity. European startups have lower survival rates and lower innovation capacity compared to US startups (Bartelsman, Haltiwanger, and Scarpetta 2004; Santarelli and Vivarelli 2007). Startups' success rates and development are to a great extent dependent on their access to capital (Fredriksen 1997, Stucki 2013). As indicated by the European Commission (2011), venture capital investments tend to be an important source of financing for innovative companies.

Venture capital investments primarily provide financing for young innovative companies with the potential for rapid growth (Prohorovs and Pavlyuk 2013). Tykvova, Borell, and Kroencke (2012) note that venture capital investments are most needed in the seed financing stage and in high-tech industries. According to a number of researchers, for example, Veugelers (2011), Reid and Nightingale (2011), Snieska and Venckuviene (2012), and Prohorovs and Jakusonoka (2012), young innovative companies have the most obvious difficulties in attracting financing at the seed stage and in case they run out of funds, they are forced to exit the market.

Providing funds to innovative SMEs is a challenge in many countries, and funding gaps are especially visible in emerging economies (OECD 2006). In the course of research based on a sample of 1600 small companies, it was determined that the growth of most firms is constrained by internal finance (Carpenter and Petersen 2002). Inadequate financing due to high volatility or absence of cash flows restrains a company from competitiveness on the market, from spending on new research, and from scaling the idea on the global markets (Moyen 2004; Kaplan, Sensoy, and Strömberg 2009). As indicated in an OECD study (2006), if an entrepreneur is not able to cover the financial gap with his own funds, the company should still aim to be self-financed and to optimize costs, which should lead to better company management.

Capital willing to accept high risks is vital for young and innovative companies due to underdeveloped sales markets, distributions channels, management systems, etc. (Fredriksen 1997). Adequate funding in the early stage of an enterprise strongly influences its future success or failure (Carter and Van Auken 1990; Gimeno et al. 1997). Researchers agree that financing at an early stage of a company's development is vital for its further existence and development.

Obviously not every entity receives capital from venture investor groups, such as venture capital funds and business angels. As Rose (2012) states, only about one company in 40 receives funds from BAs, while 1 in 400 companies receives funds from VC. Mason and Harrison (2003), referring to the European Venture Capital Association, state that only 3–4 out of 600–700 business plans are funded every year.

Business angels appear to be a very significant source of capital for young enterprises as they tend to invest in seed ventures 16 times more often than venture capital investors (Freear, Sohl, and Wetzel 1992). However, only 3% of interaction between the entrepreneur and the potential investor leads to an investment deal (Mason and Harrison 1994; Riding, Duxbury, and Haynes 1997; Sohl 2007; Maxwell 2009). So the cornerstone of the financing problem is detection of startup factors which are crucial for the investor during the selection phase—factors which provide the investor with confidence in the further successful development of the entity. The problem of discretionary financing of young innovative companies is explained by a number of aspects: the level of entrepreneurs' and management's qualifications, information asymmetry and other aspects (Yli-Renko, Autio, and Sapienza 2001; Barinova, Eremkin, and Lanshina 2015).

Knowing these criteria, entrepreneurs will be better prepared for the first round of investment from business angels or venture capital funds and as a result will have higher chances of attracting venture capital to be able to develop their startups further at a quicker pace.

Startup factors critical for early-stage investors: investors' views

Mason (2007), Sudek (2007), and Almus and Nerlinger (1999) emphasize the key role of management team members, who are essentially the founders of the venture, in business angels' decision-making. A great deal of attention is paid to leadership skills and only afterward is the product or market potential considered.

Investors value the reliability and trustworthiness of the founders greatly, while enthusiasm is important when assessing the management, as stated by Mason and Stark (2004), who believe that the people in a project are of the utmost importance as informal investors spend a lot of time with the team and the personal impression is valued greatly. Mason and Stark (2004) also note that in cases where the potential investors have expertise in the same niche as the young innovative company, their reaction to the quality of the team is less obvious than in cases where the investors lack specific knowledge in the respective economic sector. Bachher and Guild (1996) determined 95 criteria influencing the decision-making of the investor and grouped them into five key groups: (1) characteristics of the entrepreneur; (2) market specifics; (3) product or service specifics; (4) investor's requirements; (5) particular investment specifics. Freear, Sohl, and Wetzel (1992) disclose two groups of factors which might negatively influence the decision of the informal investor (e.g., business angel): attributes of the founder and of the business. The founder's attributes include lack of knowledge and experience in implementing business ideas; lack of realistic expectations, which often leads to overly optimistic forecasts and inadequate perspective assessment; personal business skills, including the inability to see the big picture and lack of desire to control the operations of the company. Business attributes include poor management skills; poor profit potential for the given risk profile; doesn't correspond to the investor's interest; insubstantial level of shareholder equity capital and lack of liquidity; insubstantial level of information disclosure, which includes an incomplete or poorly developed business plan.

As stated by Freear, Sohl, and Wetzel (1992), the market potential assessment factor has to be regarded separately, as the development of a completely new product or service for which the market is not yet ready is not an uncommon phenomenon. According to CB Insights (2014), absence of demand for a specific product is one of the most widespread reasons for startup failure.

- Clarysse, del Palacio, and Pauwels (2012) grouped the requirements claimed by venture capital investors according to the following factors: the product's potential, the market's potential, the management team and the financial potential.

Similar criteria to assess a startup company's potential for venture capital investors have been proposed by Fried and Hisrich (1994): concept, management, and profitability. The concept of the company should assume the potential for fast growth, which can be realized via substantial market growth, conquering a larger market share or a cost-cutting strategy. The concept should be based on the company's competitive edge, unless the company operates in a niche with very low competition intensity. The concept of the company should correspond with reasonable capital requirements. Evaluating the management, venture capital investors consider the past experience of the founders and key managers more than the management of the new company. Even in case of previous failures, the manager has the opportunity to demonstrate his merits. Investors obviously pay significant attention to the manager's ability to make realistic forecasts, foresee risks and proactively create a plan to tackle potential problems. Additionally, flexibility is highly valued, since it is crucial at the company's development stage.

MacMillan, Siegel, Narasimha (1985) classified factors crucial for venture capital investors into four categories: entrepreneur, venture team, product/ service and financial aspects. Roure and Maidique (1986) took the same approach in extracting similar factors, which they classified according to founders' track records, characteristics of the founding team, target market, technological strategy, and deal structure. MacMillan, Zemann, and Narasimha (1987) found that two critical factors are a winning startup team and the market growth rate, while product characteristics are of lesser importance. A similar conclusion has been made by Bruno and Tyebjee (1985), who determined that a weak management team was the main reason for negotiation failures.

Jeffrey Bussgang (2017) asked "startupers" to assess themselves according to the three most basic criteria employed by venture capital investors: the team, the market, and the business model. Bussgang also mentions the team's ability to work in an uncertain environment, pushing the limits and being creative in solving problems.

Bernstein, Korteweg, and Laws (2017), based on a sample of 4500 investors in young companies, state that the human assets of the startup are the first criteria to consider. The founding team is crucial at the beginning for differentiating the company, for developing the products and the market. However, this changes as the firm develops, hoping to attract substantial funding, and as it undergoes a standardization phase, making human assets replaceable, and creates principal business lines such as patents, technology, and physical assets, which are the key to the success (Kaplan, Sensoy, Strömberg 2009). Another study (Rea 1998) of seed-stage investors suggests that the two key factors in failing negotiations are the credibility of the business plan and the management team's competence. They have also noted that business factors are more important than product characteristics.

Interviews with more than 30 business angels in Scotland and Northern Ireland and a survey of 238 respondents allowed researchers to extract the most likely "deal killers," which turned out to be the founder and the management team (Mason, Botelho, and Zygmunt 2016). Maxwell, Jeffrey, and Lévesque (2011), based on 150 interactions between entrepreneurs and potential investors, concluded that, in contrast to the widespread understanding of angel investors' decision-making process as an evaluation and weighing of a large number of attributes, there are in fact certain fatal flaws (e.g., market potential, the financial model, relevant experience, product status) that mark the turning point when the investment is not considered further. In another study, Maxwell and Lévesque (2014) show the importance for the business angel of trust-building behavior, meaning that the entrepreneur should exhibit complete disclosure, reliance, consistency, benevolence, and alignment. A quantitative and qualitative study of Californian business angels determined that the top criteria the potential investment target should comply with are the trustworthiness of the entrepreneur, the quality of the management team, and the enthusiasm of the lead entrepreneur, while exit opportunities are also considered (Sudek 2007).

One more necessary pre-condition for the conclusion of the investment deal, which goes above the basic criteria the startup has to comply with, is intuition, the investor's gut feeling about or personal sympathy for the venture (Šimić 2015). Certain "chemistry" should arise between the entrepreneurs and the investors; therefore, Hudson and Evans (2005) call the investment decision-making process more an art than a science.

The authors of the publications mentioned in this literature review basically agree that from the point of view of investors, at the early stage of a company's development, its founders are a key asset. Therefore, a positive assessment of the entrepreneurs and the management team from investors is the decisive factor in the investment decision-making process. Additionally, venture capital investors consider the following: market size and growth rate, financial potential of the company, product characteristics.

It should be noted that the data presented in this review indicate that the specialized education and skills required for producing a product or delivering a service are not a critical factor in the decision-making process. It is also noteworthy that only one of the studies (Freear, Sohl, and Wetzel 1992) we reviewed emphasized entrepreneurs having a sufficient amount of their own financial resources as an important factor for investors.

Startup factors critical for early-stage investors: founders' views

Unlike research on the investor's side, there have been few studies inquiring about founders' views. Yet the results of such studies might be even more important for understanding whether entrepreneurs themselves acknowledge how they are assessed by potential investors and whether they try to comply with the criteria considered.

Eisenmann, Howe, and Altringer (2017), having interviewed over 140 Harvard Business School alumni who had founded technology startups, discovered that the management skills of the founders were of the utmost importance, in particular, the ability to build a founding team and lead it efficiently afterward. Moreover, it was determined that the skills a founder needs change as the venture matures, that is, founders should adapt accordingly. Additionally, Eisenmann, Howe, and Altringer (2017) emphasize the importance of skills in product design and development.

Valliere and Peterson (2007) found that experienced and novice entrepreneurs exhibit some differences in understanding the factors that influence an investor's positive decision. They also note that as entrepreneurs gain experience, they start to assign greater value to personal compatibility with investors as an important factor in choosing them.

Zobnina's (2015) field research explores the obstacles faced by Internet startup founders during each stage of development. The potential obstacles have been evaluated from both sides—the investors and the startup founders—who have identified the key issues a company faces at the pre-seed and seed financing stages. At the pre-seed stage, founders and potential investors face the same problems, which are mainly related to the idea itself and the founding team, At the seed stage, however, the problems are different. Founders see problems in product scalability, which is enhanced by lack of capital. Venture capital investors primarily see problems in management skills, the business model, or high competition, which to a great extent corresponds with the results of the majority of studies on investment criteria set by business angels and venture capital investors. Wu et al. (2009) conducted a survey of more than 200 technological startup companies' founders and discovered that for such startups, competitive advantage is determined by the founding team partners' commitments and the financial resources they possess.

It should be noted that in all four studies of entrepreneurs' points of view (Eisenmann, Howe, and Altringer 2017, Valliere and Peterson 2007, Zobnina 2015, Wu et al. 2009), the authors note the importance of the management team's quality and/or the availability of managerial skills among entrepreneurs. In two (Zobnina 2015, Wu et al. 2009) of the four studies, the authors found that, according to entrepreneurs, an important success factor for startup companies is the amount of financial resources that entrepreneurs have at their disposal. Yet only one (Freear, Sohl, and Wetzel 1992) of the more than 20 studies on investors' points of view we have reviewed noted that for venture capital investors, the amount of financial resources at entrepreneurs' disposal is an important factor for making investment decisions.

Thus, one of the discussion questions arising from the literature review is the difference in researchers' opinion on whether the amount of financial resources entrepreneurs have at their disposal could be an important factor in selecting companies for investment. To obtain an expert opinion, we conducted interviews with a number of Latvian venture capital funds, business angels, and entrepreneurs. The results of the interviews show that for venture capital funds, the amount of financial resources available to entrepreneurs is not significant, since venture capital funds are not particularly limited in financial resources, and entrepreneurs' lack of financial resources can facilitate the conclusion of an investment transaction on more favorable terms for the fund. In case a business angel is the potential startup investor, lack of financial resources on the part of the entrepreneur also strengthens the business angel's position in concluding the deal on more favorable terms. However, since the business angel is an entrepreneur himself and usually has more limited financial resources compared to a venture capital fund, the entrepreneur's financial resources are a more important factor, though still not decisive.

Entrepreneurs themselves believe that their own capital availability is an important success factor for capital attraction. This phenomenon can be explained as follows. The availability of entrepreneurs' financial resources allows them to have a stronger position in negotiations with investors, since they can continue to develop the company for some time without attracting investments and, having achieved higher development results, attract investments on more favorable terms.

Therefore, we find that there is an objective and proven difference in the views of researchers who have interviewed investors and those who have interviewed entrepreneurs on the amount of financial resources entrepreneurs have at their disposal as an important factor in their companies' selection for investment.

In concluding the literature review, we can state that our analysis of startup success factors in the capital attraction phase shows that both venture investors and entrepreneurs consider the quality of the management team and/or entrepreneurial skills to be the most important factor. This supports the administrative management theory and its postulation that an entity cannot function efficiently without properly organized management. The father of the administrative management theory, Henri Fayol, concluded that for productive staff management there should be five ways in which the management team interacts with personnel: planning, organizing, commanding, coordinating, and controlling (Fayol, 1917). This set of communication and organization processes should be the basis for efficient operations at any business entity, including a startup, which has to manage its fast development properly.

Methodology

Data

The research is based on primary data obtained through questionnaires filled out by founders of young innovative companies domiciled in Latvia and Russia. The authors first concentrated on conducting a survey questioning the company founders. Based on the primary data, the authors were able to conduct a factor analysis and a binary logistic regression analysis to determine the factors vital for seed capital attraction and their relative importance.

Forty founders (20 from Latvia and 20 from Russia) of young innovative companies were surveyed. The questionnaire was developed based on the research on startup factors influencing capital attraction. It was tested by two experienced startup founders who did not participate in the survey itself and two venture capital experts representing the Latvian Venture Capital Association. The questionnaire was comprised of 27 questions: 13 open, 7 closed, and 7 mixed. The first part (Questions 1–13) was dedicated to the organizational aspects of the company's establishment to reveal the actions undertaken to begin entrepreneurial activity and to find out whether the human capital specifics comply with the criteria vital for investors described in the literature review. The second part of the questionnaire (Questions 14–16) was meant for companies which were unsuccessful in attracting financing and focused on potential factors which appeared to be major constraints in receiving capital. The third part (Questions 17–26) was related to financing aspects such as the amount, the sources available, the actions required to attract the capital, and the factors complicating the process.

In Latvia, the target audience was founders of young innovative companies who have been appealing to the Latvian business angels' association LatBAN and to the local community Seed Forum (http://www.labsoflatvia.com/communities/seed-forum-latvia). These organizations are the largest and most important platforms for early-phase venture capital attraction in Latvia. In Russia, the main respondents were companies which have been publishing data on their projects through the tech platform Spark.ru, the business network InnMind and the website askcap.ru. Spark.ru is a web framework enabling communication between representatives of small business and enterprises. InnMind is an online platform connecting investors, experts and industry leaders with startups from the CIS and Eastern Europe and is considered to be a global ecosystem for all the participants of the innovation community. Askap.ru is an online platform that connects startup companies with investors.

The sectoral split of the respondents was as follows: 19 of 40 respondents had founded their startups in the IT industry and nine were in the energy or technological sectors. 30% of the respondents represented other industries (e.g., nanotechnology, biotechnology, e-commerce, advertising, etc.). The survey was conducted in March–April 2016.

Research design

To assess the importance of factors that affect capital attraction and to determine the weight of these factors, we divided the respondents into two groups: those who received financing and those who failed to attract capital. This approach allowed us to identify the differences in entrepreneurs' points of view on factors that affect capital attraction.

Additionally, we conducted interviews with startup sector players—entrepreneurs, business angels, venture capital funds—in order to find out whether the financial resources an entrepreneur has at his/her disposal also affect the willingness of the potential investors to invest.

Two main groups of criteria were determined in the process of comparative analysis: organizational (10 criteria) and financial (16 criteria). The comparison was made both on the level of each country and on a general level. The results of the analysis are put in the context of the literature analysis's main takeaways regarding factor importance. The most important factors, potential investors' considerations, are compared in a summary table. The factors were harmonized and logically rephrased as the designation for one factor sometimes varied from source to source. Overall the six most important factors in each group were selected based on the frequency of their mention in the literature review and on respondents' answers. Other factors have been excluded from the table due to low frequency and therefore lower importance.

Factor analysis was done based on 20 pre-selected factors with the aim of reducing the number of key factors by grouping the questions (the variables) to ease ascertainment of the factors influencing capital attraction. The analysis was conducted with the IBM software SPSS.

As a result of the factor analysis, the criteria were grouped into eight factors. The maximum likelihood factor extraction method was used, while the Varimax method was used as a factor rotation method.

After the factors were determined, the next step was assessing the influence of each factor on the ability to attract seed capital, which was represented by the dependent binary variable (attracted/did not attract seed financing). Therefore, the binary logistic regression method was used to determine the effect of independent variables on the dependent variable, which was defined as the ability of the startup to attract capital from venture capital investors.

$$p = \frac{1}{1 + e^{-z}} \tag{1}$$

, where

$$z = b_1{*}x_1 + b_2{*}x_2 + \ldots + b_n{*}x_n + c \quad (2)$$

, where B: independent variable coefficient; x: independent variable, where eight factors determined with the help of factor analysis were used.

General characteristics of the startups participating in the survey

Out of 40 companies-respondents, 15 were able to obtain seed financing. 9 companies were able to receive early-stage financing from business angels, 2 from business incubators, and 3 from venture capital seed funds. The average amount received through seed financing was €101,000. It is worth mentioning that Latvian companies received larger seed financing amounts—€136,000 per company—while Russian companies attracted a more than two times lower sum, €62,000, on average.

Overall 10 companies from Latvia and 5 companies from Russia managed to attract capital; therefore, 10 companies from Latvia and 15 companies from Russia were not successful in attracting capital. So companies from Latvia were more successful than companies from Russia; the success rate for Latvian entrepreneurs was 50%, while for Russian entrepreneurs it was only 25%.

Results and discussion

Organizational factors' importance for startups in attracting seed capital

Table 1 discloses more detailed information regarding the typology of successful companies in terms of seed financing in the context of the organizational factors.

The difference in the responses of companies which managed to attract capital and companies which failed to do so is evident on the organizational level. To almost all critical issues, such as previous experience, appropriate management, and specialized skills, higher value was assigned in the group of "successful" respondents. The results correspond to the critically important investment factors mentioned in the majority of studies based on interviews with business angels and venture capital investors (Tyebjee 1985; Mason, Botelho, and Zygmunt 2016; Bernstein, Korteweg, and Laws 2017). The results obtained also confirm the ongoing topicality of the administrative management theory put forward by Fayol (1917): the answers of the respondents indicate entrepreneurs' belief that efficient management is a necessary prerequisite for a successful entity's development and, therefore, for attraction of a potential investor. However, at the earliest company development stage, administrative management skills and

Table 1. Comparison of the organizational factors' relevance for startups in attracting seed financing.

No.	Organizational factors	Attracted seed financing (%)	Did not attract seed financing (%)
O1	Previous experience in establishing companies	87	44
O2	Management skills of the team, including the founder	93	76
O3	Specialized education and skills required for producing the product or delivering the service	93	84
O4	Option pool	40	36
O5	Use of a consultant's/mentor's help	27	48
O6	Use of a business accelerator's services	33	20
O7	Use of managerial support	53	44
O8	Use of services for provision of the premises	33	36
O9	Use of business consultancy	13	24
O10	Use of business incubators' services	13	28

experience appear to be less important compared to the entrepreneurial characteristics and spirit of the founders. But as the company develops, the role of these factors (planning, organizing, commanding, coordinating, controlling), put forward by Fayol in the administrative management theory, becomes more important for the success of the company; therefore, investors, in considering development prospects, assign to this factor (assessment of the quality of the management team and the founders) the most tangible value. It is a widespread phenomenon that at a certain point in a company's development, the founders' management skills lag behind the required level and the founders themselves, perhaps on the basis of a proposal from investors, hire experienced managers who possess the required set of skills to bring the company forward. Our findings correlate with Hellmann and Puri (2002), who determined that companies which have attracted venture capital investments are likely to replace the founder with an outside CEO.

It is noteworthy that help from mentors and business consultants as well as an incubation period in a business incubator were clearly not decisive factors in the process of seed capital attraction. However, the results of the survey indicate that accelerators and managerial support were helpful when persuading investors to invest in a startup. The frequency of answers regarding the existence of an option pool for additional motivation of key employees was not significantly different among the groups of enterprises, so it does not exert significant influence on the success of capital attraction.

Comparing the responses from Latvia and Russia, one finds similar results: previous experience and management skills make all the difference (Figure 1).

In both countries, in case the founders have previous experience in building a company, they have a higher probability of attracting financing. With regard to the professionalism of the team, the results for capital

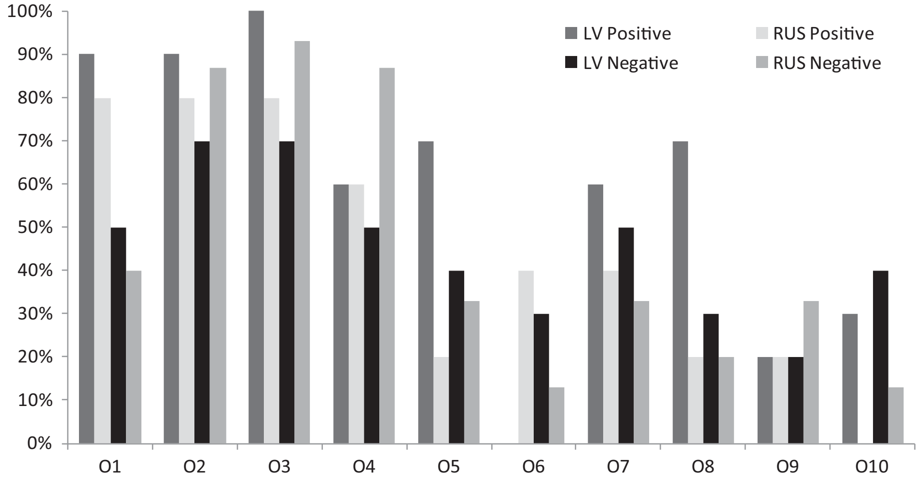

Figure 1. Comparison of the organizational factors' relevance for startups in attracting seed financing in Latvia and Russia (positive and negative experience in capital attraction).

attraction in Russia seem counterintuitive, as companies with more professional teams were less successful in the seed financing round. These results might be explained by an overestimation bias on the part of the respondents when evaluating the founding and management teams' professionalism.

The results of the survey in Latvia and Russia provide a clear picture of the frequency of using consulting services, allowing us to conclude that Latvian entrepreneurs were seeking external help and support when building a startup more actively than their Russian peers. The services that did not add tangible value in Latvia were incubators, accelerators, and business consultancy. This might be explained by the still underdeveloped culture of supporting startup infrastructure. The answers provided by the Russian startup entrepreneurs do not allow us to make robust conclusions on which organizational aspects were helpful, but it can be seen that business accelerators and managerial support provided a certain boost in the seed capital-seeking phase.

To conclude the section on organizational factor importance, it should be noted that startup founders, regardless of their success in the seed capital attraction phase, mentioned (1) the management team and entrepreneurial skills and (2) education and experience in the production and supply of products/services as the most important factors. Entrepreneurs who managed to attract seed financing assigned a higher value (by 17% points) to the management team and entrepreneurial skills factor and a higher value (by 9% points) to the specialized education and skills factor compared to entrepreneurs who did not manage to attract seed financing.

Entrepreneurs who succeeded in attracting seed financing ranked previous experience in establishing companies at third place and gave this factor

Table 2. Comparison of the financial factors' relevance for startups in attracting seed financing.

No.	Financial factors	Attracted seed financing (%)	Did not attract seed financing (%)
F1	Investor's doubts about the company's financial objectives	13	32
F2	Disagreement about the required investment amount	13	24
F3	Insufficient market size	27	20
F4	Uncompetitive business model	0	20
F5	Incompetence of the investor	13	0
F6	Lack of a finished product;	20	48
F7	Investor's doubts about the professionalism of the team	27	32
F8	Investor's doubts about the product	27	32
F9	Disagreement about the investor's share	13	24
F10	Lack of export orientation	0	8
F11	Other factors	20	4
F12	Participation in business angels' sessions	20	24
F13	Direct contacts with business angels;	73	44
F14	Direct communication with investors;	60	64
F15	Seeking assistance in the search for investments	13	40
F16	Appeal to venture capital funds	40	44

a high score (87%), only 6% points lower than the response frequency for the main factor. Among entrepreneurs who failed to attract seed financing, the response frequency for this factor was only 44%, positioning this factor at fourth place in the ratings of entrepreneurs who failed to attract capital. However, the majority of studies confirm that investors attach great importance to this factor. The differences in the response results to a certain extent show that entrepreneurs who managed to obtain seed investment from venture capital investors complied more substantially with the basic criteria that guide business angels or venture capital funds in making investments.

Financial factors' importance for startups in attracting seed capital

Considering the answers regarding startups' financial aspects, one can see that with the exception of direct communication with investors and business angels, financial factors overall are less important compared with organizational factors. It is also worth mentioning that evidence of the importance of financial factors is very slight in the scientific literature. The majority of sources mention failure in the business plan or market growth as the key reason why the deal does not happen (MacMillan, Zemann, and Narasimha 1987; Rea 1998; Maxwell, Jeffrey, and Lévesque 2011).

Table 2 demonstrates that investors, according to entrepreneurs, usually pay attention primarily to the readiness of the company for capital inflow, such as the stage of product development and competitiveness of the product and business model. In case the investor's assessment of the investment project does not comply with the view of the company's management (e.g.,

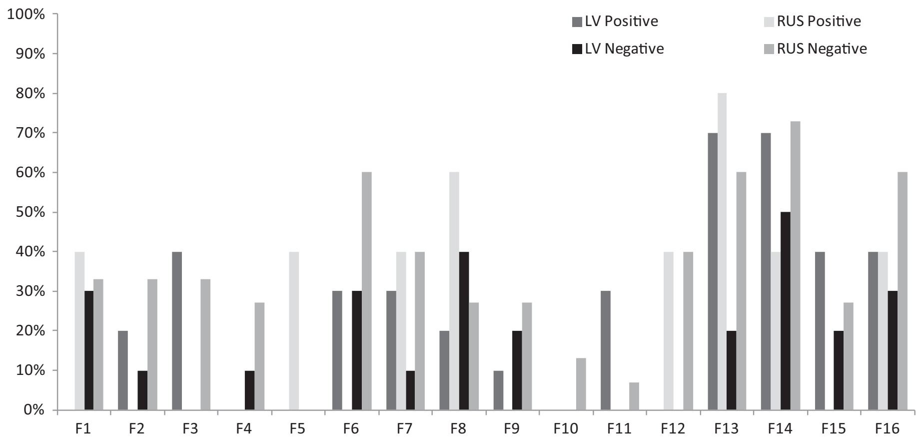

Figure 2. Comparison of the financial factors' relevance for startups in attracting seed financing in Latvia and Russia (negative and positive experience in capital attraction).

disagreement on the amount of the investment and financial targets), this might also become a serious obstacle to attracting capital.

Over 73% of companies which reported a successful seed financing stage indicated they had direct communication with business angels. This factor plays a more significant role than participation in the sessions with business angels, having contact with other investors or seeking investments from venture capital funds.

In comparing the answers provided by entrepreneurs from Latvia and Russia (Figure 2), it should be noted that entrepreneurs from Latvia are less likely to run into disagreement with the investor about the amount of capital requested. This might indicate more adequate assessment of companies in Latvia. Lack of finished products and investors' doubts about the professionalism and the competitiveness of the product often prevent young innovative Russian companies from attracting seed financing. These three factors indicate that Latvian entrepreneurs have a better ability to persuade an investor and present him with information in the most advantageous way when seeking capital.

None of the Latvian companies questioned participated in business angels' investment sessions, unlike Russian entrepreneurs. Russian companies did participate, but from the response rates one can conclude that it obviously was not a decisive factor when seeking capital. What was more important in both countries according to the data obtained was direct communication with business angels, particularly in Latvia.

Startup founders in both countries actively engaged in direct communication with potential venture investors, which obviously benefitted Latvian entrepreneurs, but this was not a decisive factor in the case of the Russian startups. Startup founders in both Russia and Latvia appealed to venture funds and sought assistance in their search for capital. According to the survey results, Russian entrepreneurs appealed to venture funds more often than startup founders in Latvia, yet they were not as successful in attracting venture capital investments. Based on the results obtained, one might conjecture that Russian entrepreneurs, despite a good network of potential investors, have trouble offering them attractive opportunities, whether due to poorly developed business concepts or poor presentation skills, or because they struggle to build trusting relationships with them. It could also be that Russian entrepreneurs are less compliant with the criteria imperative for investors or the specifics of the Russian venture capital market.

The evidence of the importance of financial factors provided in the scientific literature mainly concentrates on the plausibility of the business plan and market characteristics such as competition intensity, market size, and growth, while product characteristics appear to be less important. The analysis of the survey results indicates that business plan factors (as proxied by disagreement about financial objectives or the investment amount) are rather important, as a third of the respondents who did not manage to attract capital mentioned it as an obstacle. The stage of product preparation is also important as 48% of entrepreneurs who were unsuccessful in the capital attraction phase noted it as a barrier. So the survey results support the scientific evidence with regard to business plan factors, but do not prove the importance of the market size factor.

Startup factors' importance: scientific evidence vs. survey data

Table 3 summarizes the main findings of the literature review as well as our survey results, providing an overview of the most important factors in the capital attraction phases from investors' and founders' perspectives.

As shown in Table 3, of the six most important factors, only the first factor—the management skills of the founder and the team—is equally important according to the literature review, from the point of view of both investors and of entrepreneurs, and according to the results of our survey. In our survey, entrepreneurs from Latvia and Russia did not mention factors necessary for attracting venture capital investments—such as market size and growth rate, competition, product characteristics and scalability, and the founder's trustworthiness and reliability—among the first six factors. Underestimation by entrepreneurs of such factors may be one of the reasons for difficulties in attracting venture capital as well as an indicator of venture entrepreneurship's underdevelopment and limited experience in venture investments in Latvia and Russia.

Table 3. Summary table: comparison of the six most important startup success factors in attracting capital according to the scientific literature and the research results, sorted according to importance.

Startup success factors according to investors (literature evidence)	Startup success factors according to entrepreneurs (literature evidence)	Startup success factors according to entrepreneurs (survey results)
• Management skills of the founder and the team; • Market size and growth rate, competition; • Financial potential (business plan, profitability of the project, exit opportunities); • Product characteristics and scalability; • Founder's trustworthiness and reliability; • Founder's previous experience;	• Management skills of the founder and the team; • Product characteristics and scalability; • Entrepreneurs' capital availability; • Market size and growth rate, competition; • Founder's trustworthiness and reliability; • Marketing and sales skills.	• Management skills of the founder and the team; • Specialized education and skills; • Founder's previous experience; • Direct communication with business angels and investors; • Managerial support; • Financial potential (objectives, investment target).

Results of the factor analysis

To determine which factors exert the most significant influence on attracting seed investments, the authors selected the organizational and financial variables analyzed above and enriched this selection with the pre-seed financing amount, the number of pre-seed financing sources and the length of the period from idea generation to attracting initial investments (Table 4).

Factor analysis was conducted to understand the interrelations between the variables, their possible grouping and their common influence on pre-seed investment attraction. The KMO Test is 0.437 (Table 5), while the recommended value is above 50%. In this case, a thorough analysis of the results obtained is required. Bartlett's test significantly exceeds one (Table 5), indicating strong correlation relationships.

Overall there are 8 factors that explain 62% of the total variance (Table 6), which exceeds the recommended value of 50%. The rotated factor matrix shows 8 distinct factors, which group pre-selected variables (Table 7). A detailed description and the names assigned to the factors determined through the factor analysis are provided in Table 8.

Results of the regression analysis

The variables integrated into the 8 factors are extracted from the answers of the startups' founders or management and should, therefore, be regarded as their point of view and self-evaluation when attracting the capital. In order to understand the influence of the factors on young innovative companies' probability of obtaining financing, the method of binary logistic regression was used. The factors used as the independent variables were as follows: F′1: Professional Help; F′2: Networking; F′3: Investor's Evaluation

Table 4. Description and symbols of criteria for factor analysis.

Criterion symbol	Criterion description
O1	Previous experience in establishing companies
O2	Management skills of the team, including the founder
O3	Specialized education and skills required for producing the product or delivering the service
AD1	Period from idea generation to the first investments in the company
AD2	Number of shareholders prior to seed investments
O4	Option pool
F1	Investor's doubts about the company's financial objectives
F2	Disagreement about the required investment amount
F3	Insufficient market size
F6	Lack of a finished product;
F7	Investor's doubts about the professionalism of the team
F8	Investor's doubts about the product
F9	Disagreement about the investor's share
O5	Use of a consultant's/mentor's help
O6	Use of a business accelerator's services
O7	Use of managerial support
O8	Use of services for provision of the premises
O9	Use of a business consultancy
O10	Use of business incubators' services
AD 3	Amount of seed capital attracted

O: organizational criteria; F: financial criteria; AD: additional criteria.

Table 5. Factor analysis output: KMO and Bartlett's test, rotated factor matrix.

Kaiser–Meyer–Olkin measure of sampling adequacy		0.437
Bartlett's test of sphericity	Chi-Square	255.75
	df	190
	Significance	0.001

of the Project; F′4: Basic Resources; F′5: Preparedness of the Company; F′6: Management Motivation; F′7: Market Size; F′8: Management Skills. It should be noted that the multicollinearity problem is not relevant as factor analysis allowed us to extract factors which do not exhibit mutual correlations. The classification accuracy of the model is very high, at 76%, and Nagelkerke R^2 is 0.448, which indicates a good model fit (Table 9).

The regression analysis indicates that F′3 (Investor's Evaluation of the Project) turned out to be significant when predicting the probability of attracting capital. With a lower significance factor, F′8 (Management Skills) can also be included in the prediction equation (Table 9).

When running the regression with F′3 and F′8 as the two independent variables, the prediction power decreases but is still at the acceptable level of 62.2%. The Nagelkerke R^2 of the regression model is 30.5% (Table 10). The Hosmer–Lemeshow test's significance, indicating the goodness of the model's fit, is at 0.94, which shows that reducing the number of factors to the most significant ones increases the model's fit. The negative beta coefficient of F′3 is explained by its constituents, as they have a primarily a negative connotation: the investor's disagreement and doubts.

The results obtained support the evidence provided in the scientific literature from the point of view of both investors and startup founders. Indeed, the management factor seems to be critical, while financial aspects

Table 6. Factor analysis output: total variance.

	Initial eigenvalues			Extraction sums of squared loadings		
Factor	Total	Variance	%	Total	Variance	%
O1	3.278	16.388	16.388	2.146	10.730	10.730
O2	2.670	13.349	29.737	1.706	8.530	19.260
O3	2.123	10.614	40.351	2.494	12.469	31.729
AD1	1.869	9.345	49.696	1.481	7.407	39.136
AD2	1.629	8.145	57.841	1.501	7.506	46.642
O4	1.414	7.071	64.913	1.128	5.642	52.284
F1	1.180	5.898	70.811	1.146	5.728	58.012
F2	1.044	5.219	76.030	0.845	4.225	62.237
F3	0.794	3.968	79.998			
F6	0.691	3.455	83.453			
F7	0.633	3.166	86.619			
F8	0.551	2.754	89.373			
F9	0.483	2.413	91.786			
O5	0.423	2.113	93.898			
O6	0.372	1.859	95.758			
O7	0.275	1.373	97.131			
O8	0.208	1.042	98.173			
O9	0.162	0.809	98.981			
O10	0.130	0.650	99.631			
AD3	0.074	0.369	100.000			

Table 7. Factor analysis output: rotated factor matrix.

	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8
O10	0.825							
O9	0.773							
O8	0.701							
O5	0.580							
AD1	0.486							
O7		0.946						
AD2		0.434						
F1		−4.19	0.624					
F2			0.616					
F9			0.575					
AD3			0.479		0.415			
F8				0.859				
F7				0.621				
O6				0.402				
F6					−0.677			
O1					0.597			
O4						0.899		
O3						0.469		
F3							0.845	
O2								0.773

Extraction method: maximum likelihood; rotation method: Varimax with Kaiser normalization.
Rotation converged in 10 iterations.

such as the required investment amount or the company's financial objectives are also important for attracting capital in the seed stages (Tyebjee 1985; MacMillan, Zemann, and Narasimha 1987; Rea 1998; Sudek 2007; Maxwell, Jeffrey, and Lévesque 2011; Mason, Botelho, and Zygmunt 2016; Bernstein, Korteweg, and Laws 2017; Bussgang 2017; Eisenmann, Howe, and Altringer 2017).

Other factors, such as preparedness of the company or networking skills, which seem intuitively to be important, do not have any effect when a

Table 8. Factors determined as a result of factor analysis.

Factor title	Decription of the factor
F′1: Professional Help	The first factor is comprised of the five variables which are primarily related to professional support: business incubators' services, business consultancy, premises provision, a consultant's/mentor's help, and the length of the period after forming the business idea till the first investments are received.
F′2: Networking	The second factor includes managerial support from friends, acquaintances etc. and the number of shareholders prior to seed investments. The unifying element is networking; that is, establishing a company proceeds more smoothly in case of a large network.
F′3: Investor's Evaluation of the Project	The third factor is comprised of variables which are related to the investor's assessment and decision-making process: the investor's doubts about the company's financial objectives, disagreement about the required investment amount, and disagreement about the proposed share and the amount of seed capital attracted.
F′4: Basic Resources	The fourth factor is also related to the investor's belief in the company and the team. It includes the following variables: the investor's doubts about the product and the professionalism of the team and the use of a business accelerator's services. The team and the product are considered to be the basic resources for successfully starting a company.
F′5: Preparedness of the Company	The fifth factor is comprised of the aspects indicating the preparedness of the company for the next financing round: availability of the finished product and previous experience in establishing companies. Both factors offer a hint about how ambitious and reliable the team is and indicate the company's level of preparedness for further development and attraction of professional investors.
F′6: Management Motivation	The sixth factor is to a great extent related to the management's motivation (option pool) and skills and specialized education relevant for successfully producing and launching the product.
F′7: Market Size	The seventh factor is comprised of just one variable indicating the potential market size (insufficient market size).
F′8: Management Skills	The eighth factor includes, perhaps, the most important ingredient when establishing a startup company: the management skills of the team and the founder(s).

Table 9. Summary results of binary regression model.

	B	SE	Significance	Exp(*B*)
FACT_1	−0.715	0.588	0.224	0.489
FACT_2	0.233	0.461	0.613	1.262
FACT_3	−2.612*	1.246	0.036	0.073
FACT_4	−0.129	0.525	0.806	0.879
FACT_5	0.459	0.564	0.416	1.582
FACT_6	0.565	0.569	0.321	1.760
FACT_7	0.373	0.520	0.473	1.452
FACT_8	1.151**	0.699	0.100	3.162
Constant	−1.207	0.597	0.043	0.299

*Significance at 5%.
**Significance at 10%.
Model summary: Cox & Snell R^2 0.329, Nagelkerke R^2 0.448, Hosmer–Lemeshow test: 12.32 (Significance 0.138).
Model prediction power: 75.7%.

young innovative company seeks capital. The calculated statistics of the regression show that less than 50% of the dependent variable's dispersion is explained. The primary reason for such a low coefficient is the limited number of respondents. It should also be taken into account that the

Table 10. Summary results of binary regression model including Factor 3 and Factor 8.

	B	SE	Significance	Exp(B)
FACT_3*	−1.779	0.879	0.043	0.169
FACT_8	0.810	0.514	0.115	2.249
Constant	−0.857	0.472	0.069	0.424

*Significance at 5%.
Model summary: Cox & Snell R^2 0.224, Nagelkerke R^2 0.305, Hosmer–Lemeshow test: chi-square 2.835 (Significance 0.944).
Model prediction power: 62.2%.

companies which did not attract financing outnumber the companies which were successful in obtaining capital.

Conclusions

In researching the organizational and financial reasons for failing to attract seed capital, the authors of the present study have analyzed the reasoning of the founders startups, having questioned 40 founders (15 managed to attract capital, 25 failed) of young innovative companies in Latvia and Russia.

Based on the analysis conducted, it was concluded that companies which tend to be more successful in attracting seed funding are managed by entrepreneurs who have had previous experience in creating business entities and are capable of building a team with employees who have appropriate experience, specialized education and high-level management skills. The most common obstacle in the capital attraction process is lack of professionalism of the team, which confirms the evidence found in the scientific literature.

The results we obtained based on the review of empirical studies and on a survey of startup founders testify that the provisions of the theory of administrative management developed by Fayol at the beginning of the last century constitute the primary criteria venture capital investors take into account when assessing startup companies.

Confirmation of the questionnaire responses was received through conducting a factor and regression analysis, which determined that the investor's evaluation of the project and the team's management skills are the most significant factors in predicting success in attracting financing.

The expert interviews with venture capital investors and startup founders revealed the objective contradiction: Founders' capital availability does not appear to be significant for venture capital investors, while entrepreneurs believe it is important, which is explained by the power this factor provides in investment deal negotiations. The present research has its limitations as the responses of 40 startup founders may not be enough to ensure complete reliability of data. Perhaps the results obtained might differ from the evidence in the scientific literature as startup companies and investors may have certain specifics in Latvia and Russia, especially taking into account that experience in venture capital investing has not yet accumulated in Central and Eastern European countries to the extent that it has accumulated in North America or Western Europe, so that the trend of investing in startups is relatively underdeveloped. Our findings can form a basis for further studies on the viewpoints of founders operating in other countries of Western Europe and Central and Eastern Europe as well as in other transition and post-transition economies.

Recommendations for further research

The present research could be further expanded geographically to detect differences between countries and regions from both investors' and entrepreneurs' perspectives. Additionally, it would be worth looking into factor importance according to the type of sector where a particular startup operates.

Due to the very limited number of studies on the success criteria for attracting seed investments from founders' perspectives, it is necessary to expand the research in this area. In conducting this research, it would be advisable to divide the founders who participate in the survey into two groups—those who received financing and those who failed to obtain it—as the viewpoints of the two groups of founders regarding factors and their weight in influencing successful capital attraction might differ significantly. When conducting research on such factors, one should take care to rank them separately for each company's development stage in order to get plausible results, as the factors themselves and their weights will be different depending on the company's stage: pre-seed, seed or more mature development stages.

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